

JUNE 2024

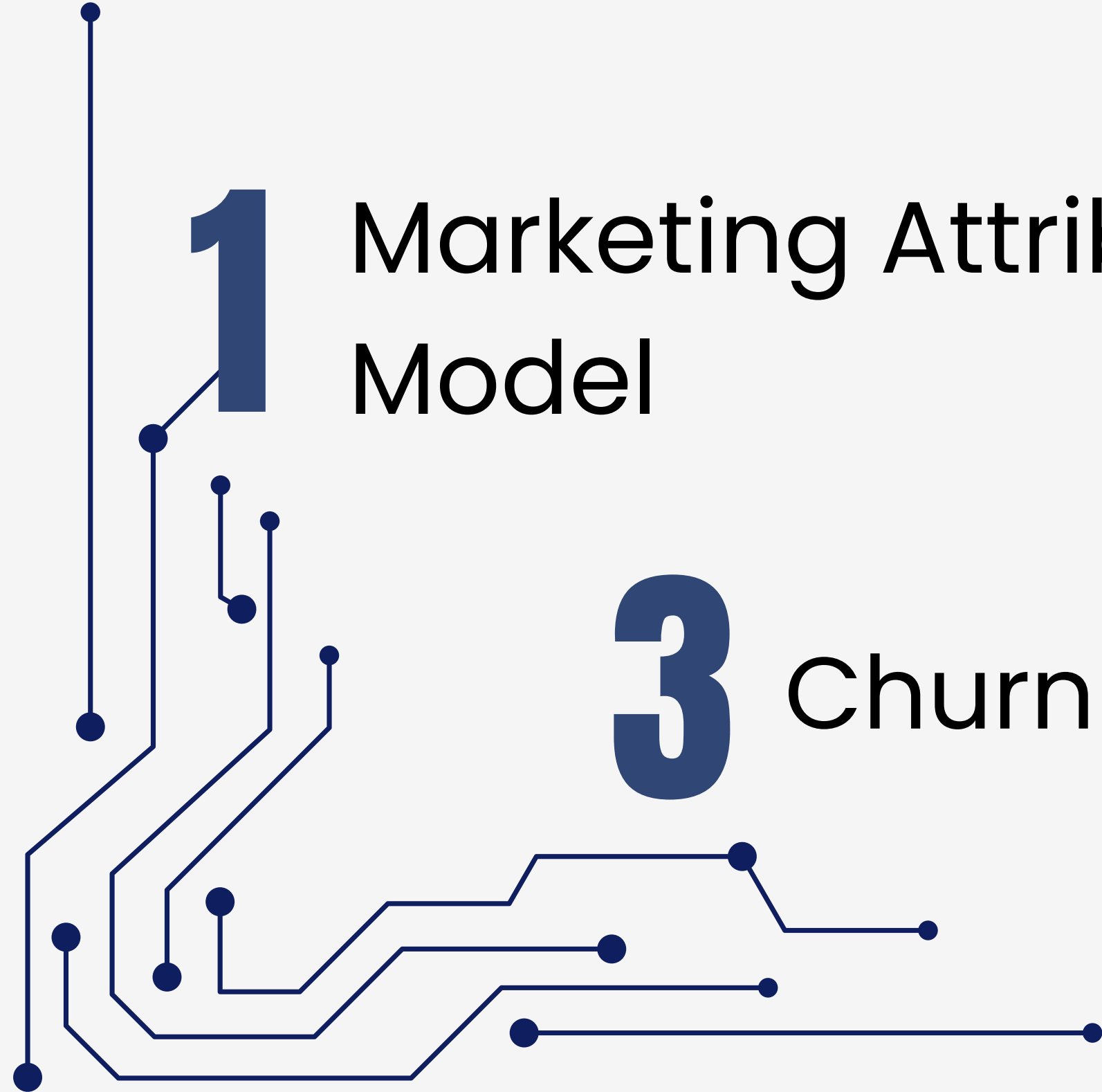
DIGITAL MARKETING RESEARCH

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USE CASES

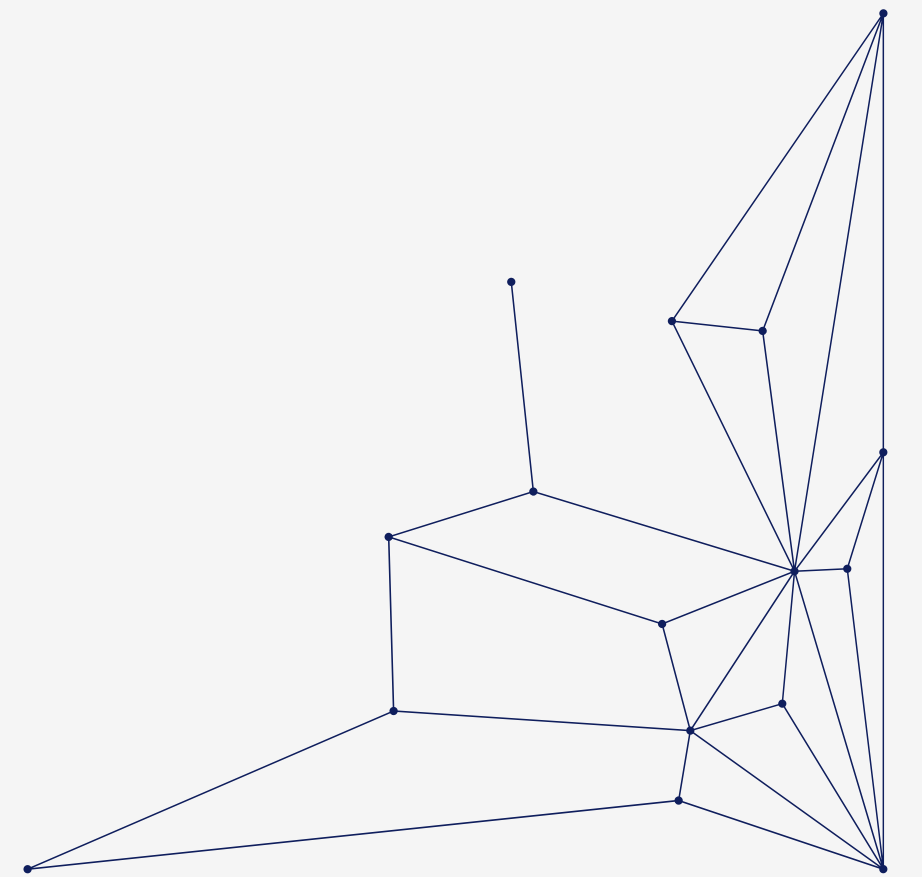
studied



1 Marketing Attribution Model

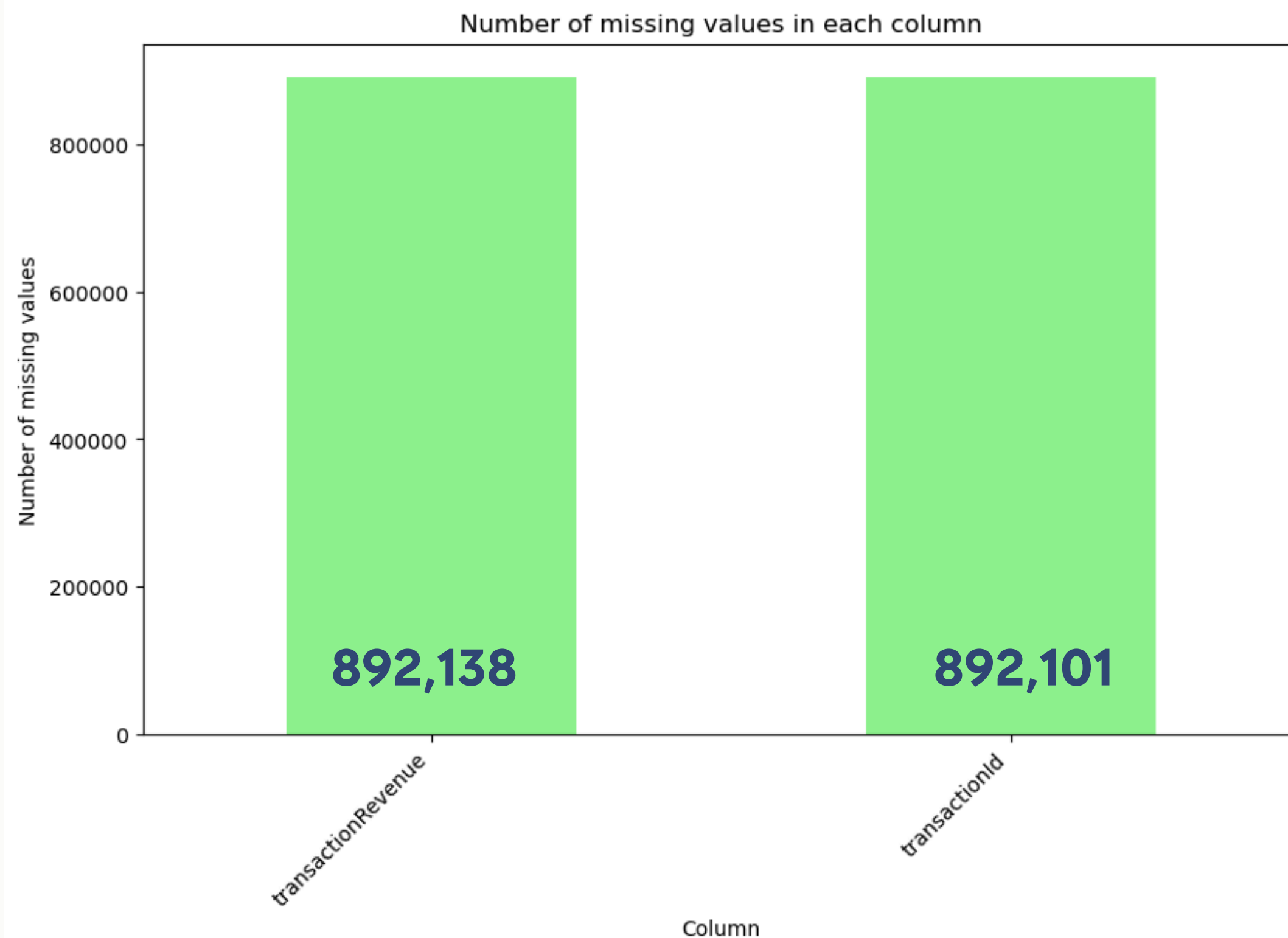
2 Customer Segmentation

3 Churn Rate Forecasting



1/ Marketing Attribution Model

Data cleaning process

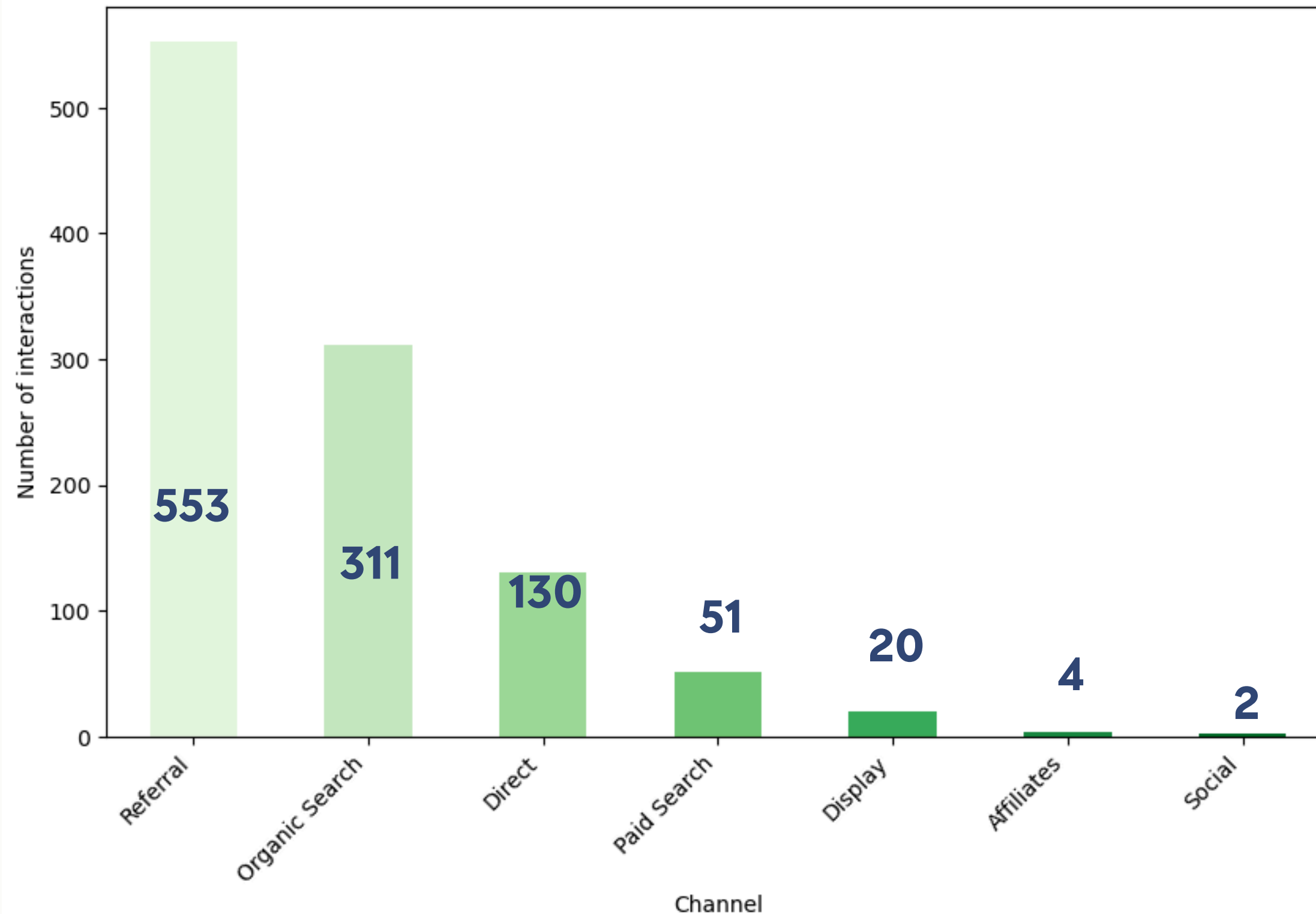


'Date' column's cleaning process

Dataset is filtered to include only the last 30 days, based on the maximum date in the dataset.

Minimum date in the dataset:
2016-08-01
Maximum date in the dataset:
2017-08-01

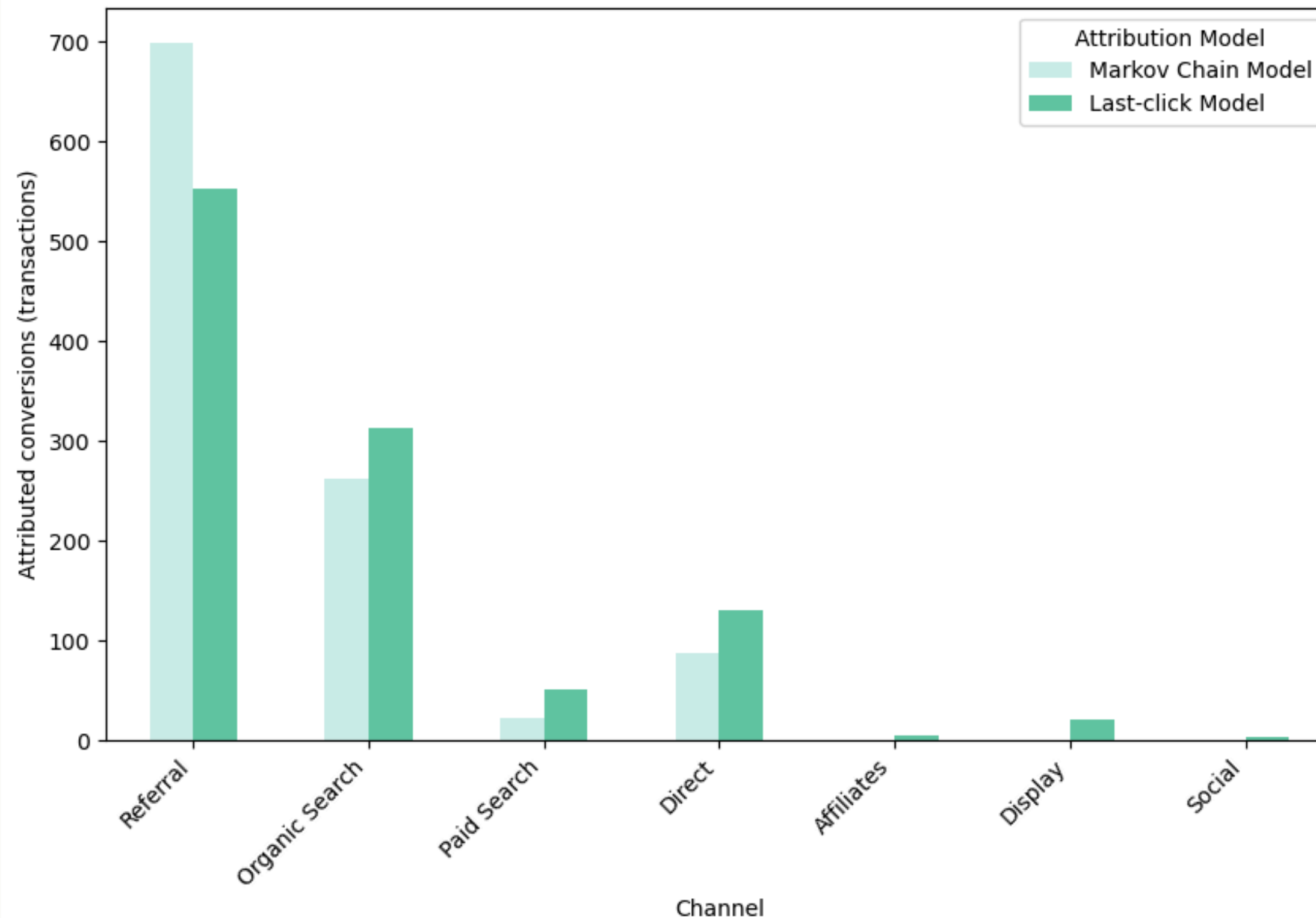
Interaction volume by channel



Last-click Model Attribution vs Markov Chain Attribution



Last-click Model Attribution vs Markov Chain Attribution



2/ Customer Segmentation

Variables	Description	Types
date	Purchase Date	datetime64[ns]
CustomerId	Unique Customer ID	object
InvoiceId	Unique Transaction ID	object
productPrice	Price per unit of product	float64
productQuantity	Quantity of products bought	int64

RFM analysis

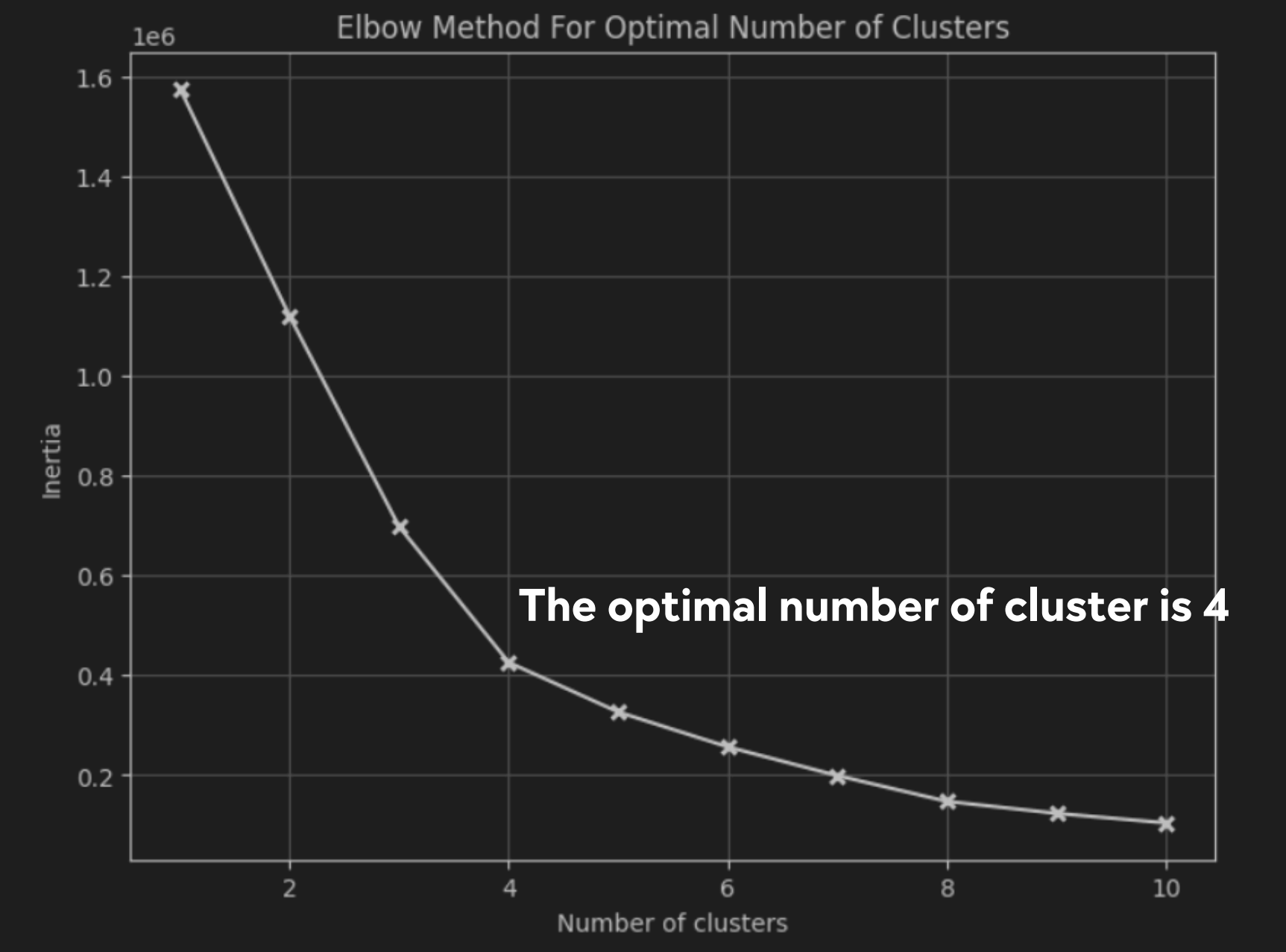
11 rows ▾ 525342 rows × 3 columns pd.DataFrame ↗			
CustomerId	Recency	Frequency	Monetary
+++SJgx/2IJ+dXq7vF8C0g==	124	1	107.85
+++aKiAiXhTfaqCLC/kyWA==	381	1	11.98
++/G67YHZTMKdpvANeYPLw==	442	1	33.54
++/GTDXvJzF11ZIUz81SPg==	217	1	41.94
++0Dxza60/nPDbf0RBYuuA==	34	1	53.88

R

F

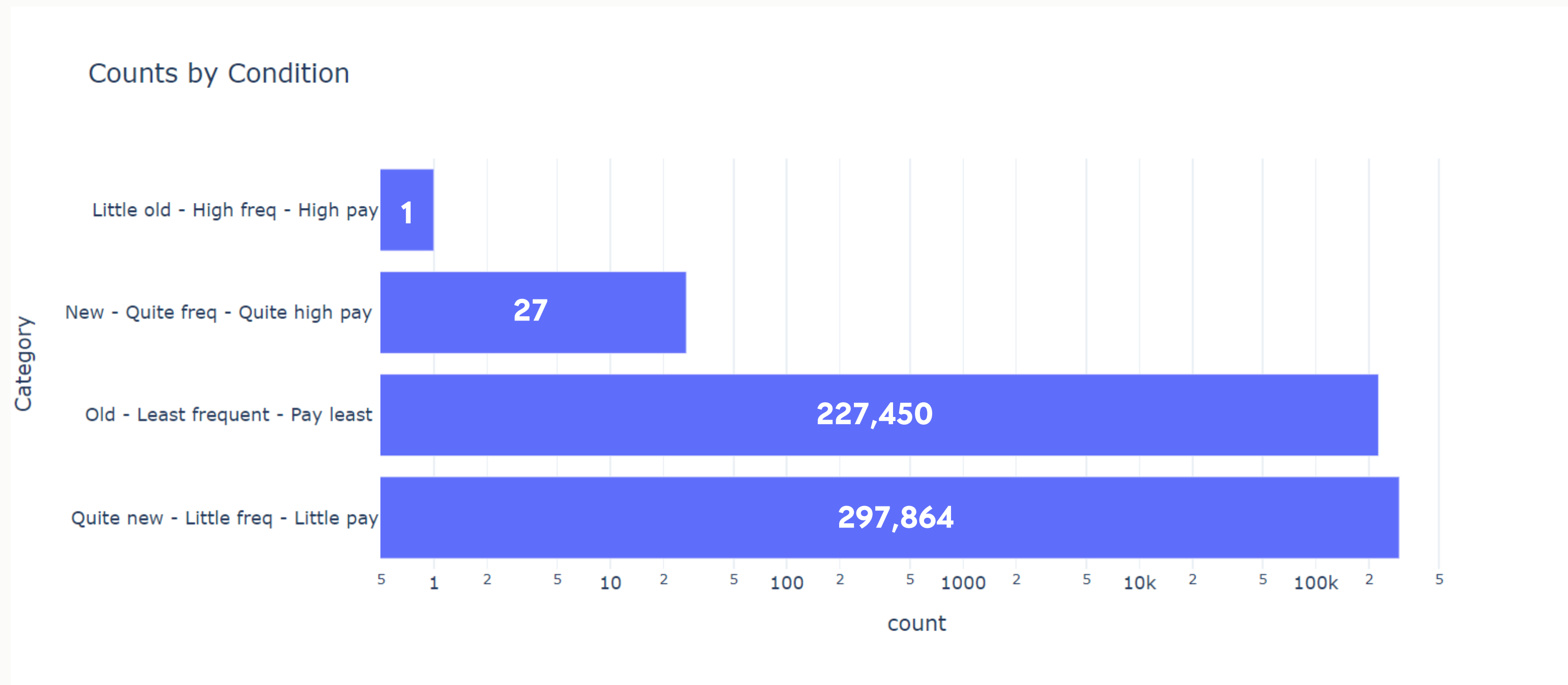
M

Clustering customers: kmeans

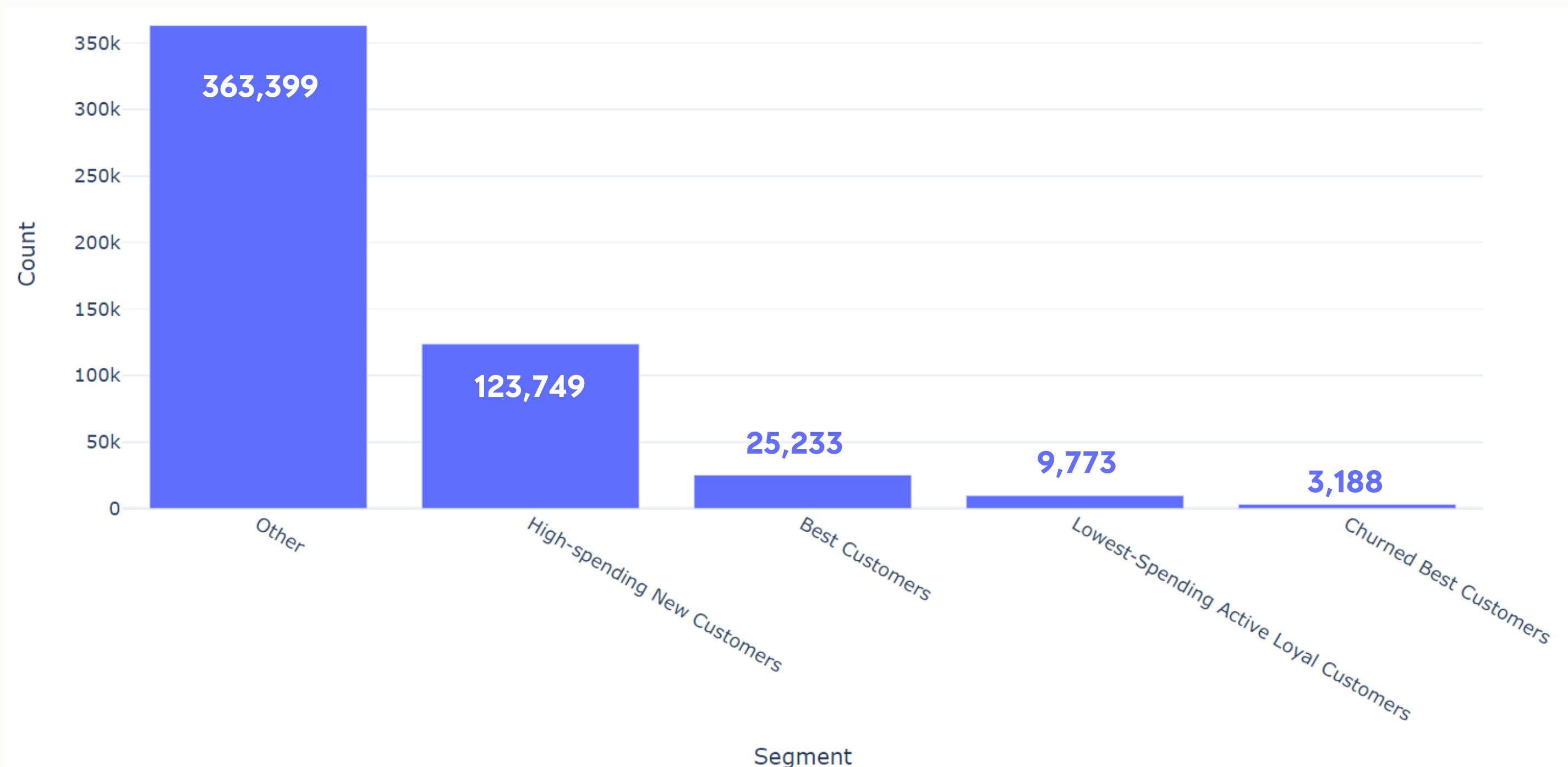


Clusters	R	F	M
1	354.67	1.16	1.7e+02
2	106.52	1.81	8.8e+02
3	123.00	1676.00	4.0e+07
4	53.93	518.33	2.6e+06

Clustering customers: kmeans

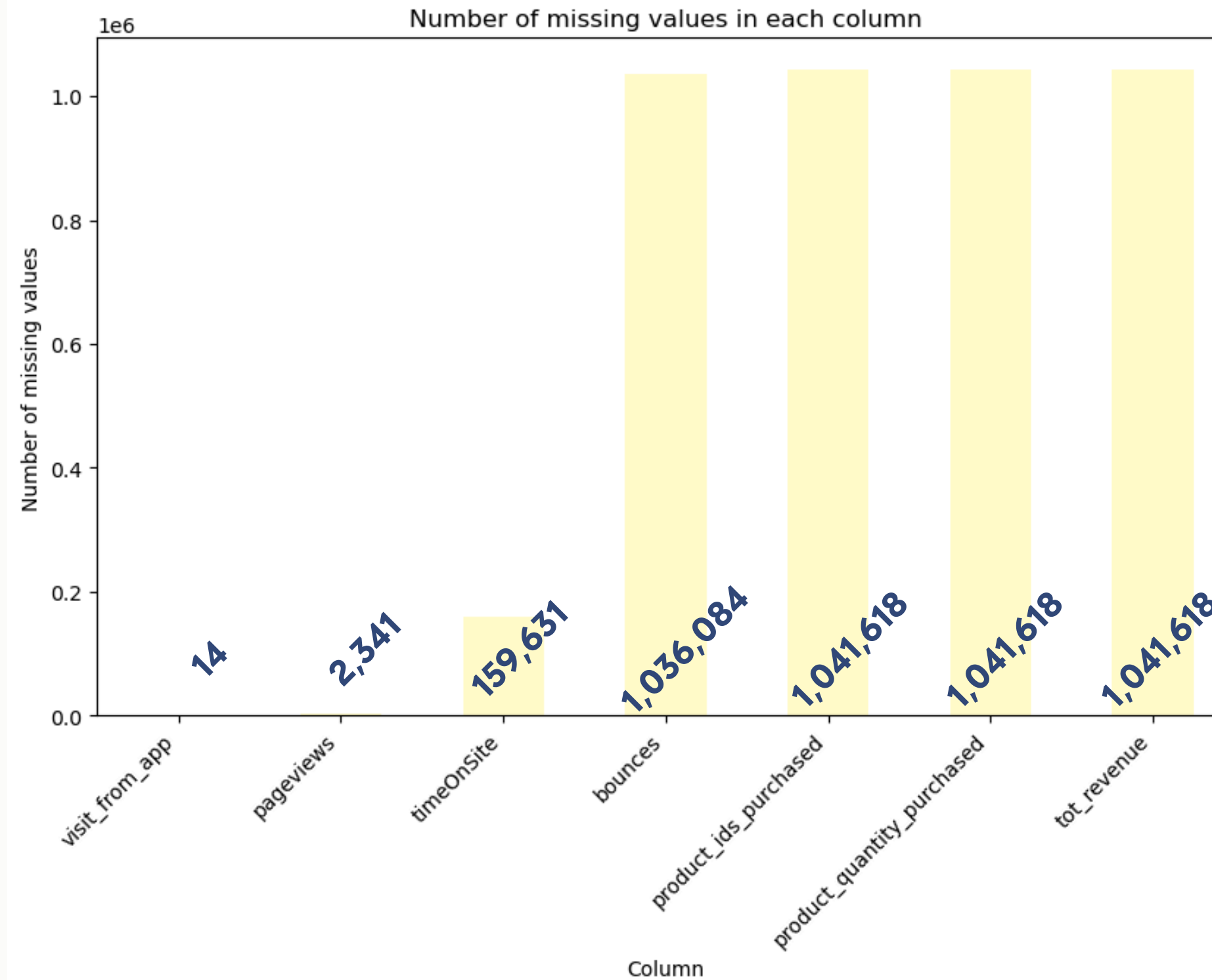


Clustering customers by tiers



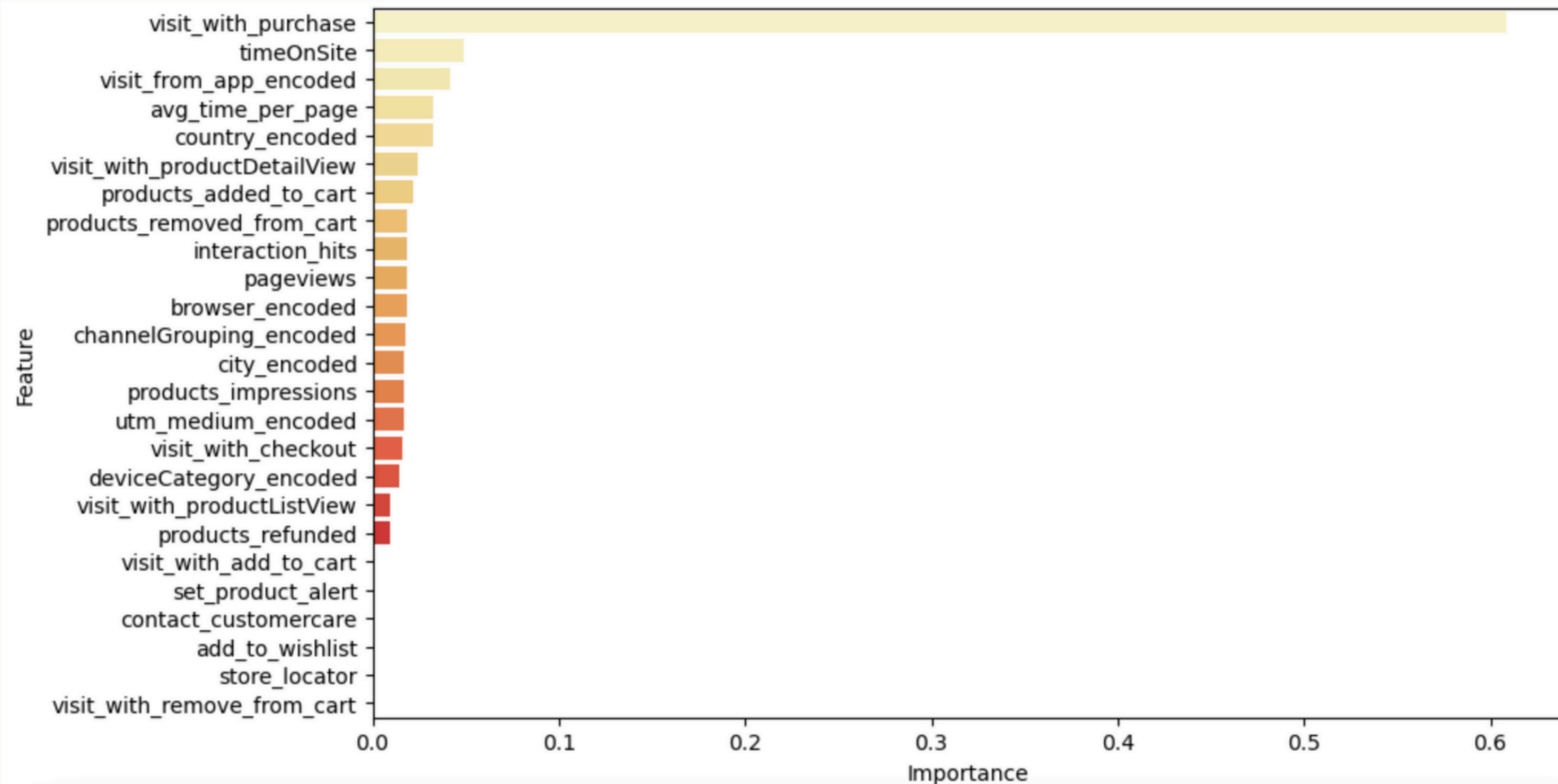
3/ Churn Rate Forecasting

Data cleaning process



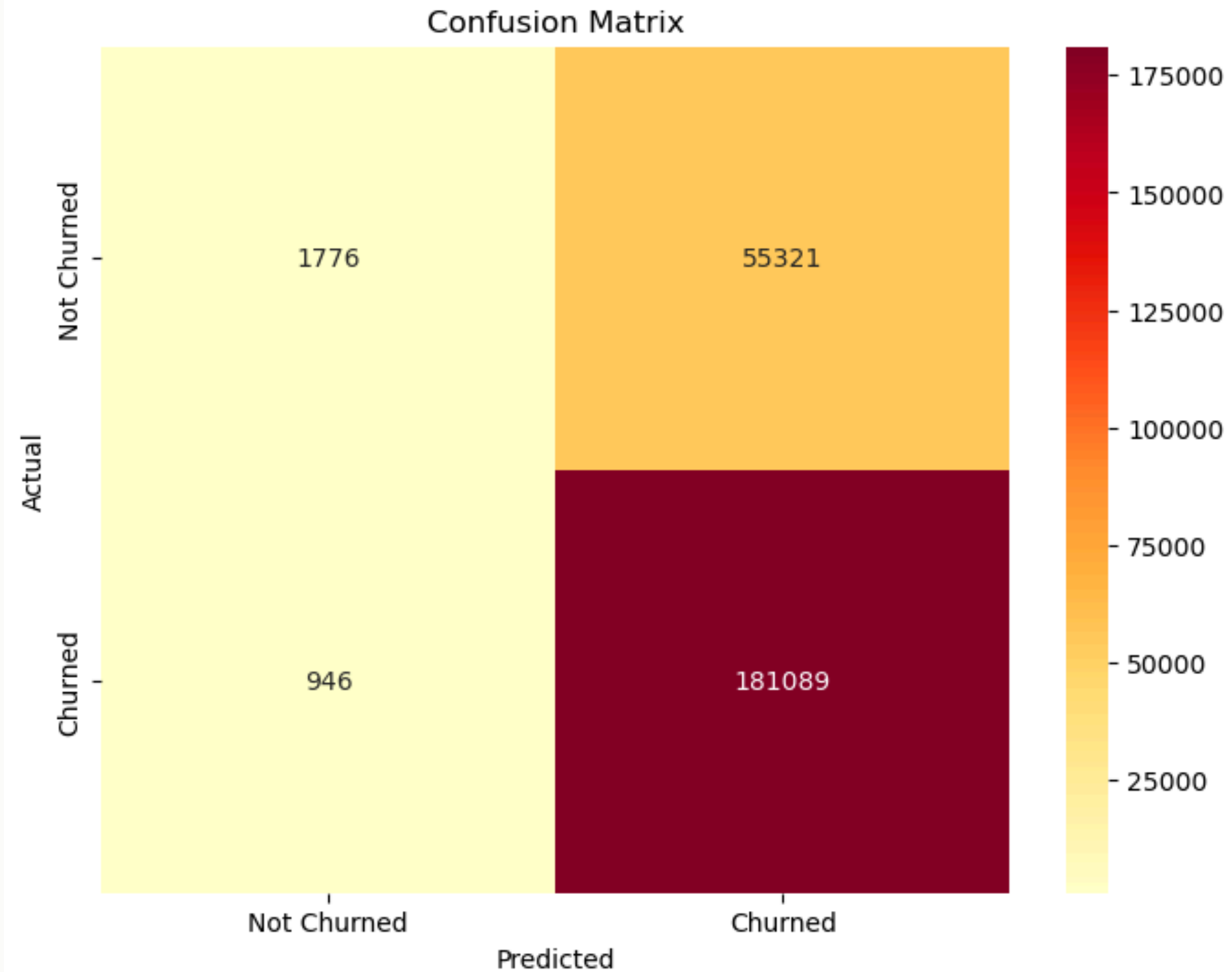
XGBoost model's feature importance

Including newly created features

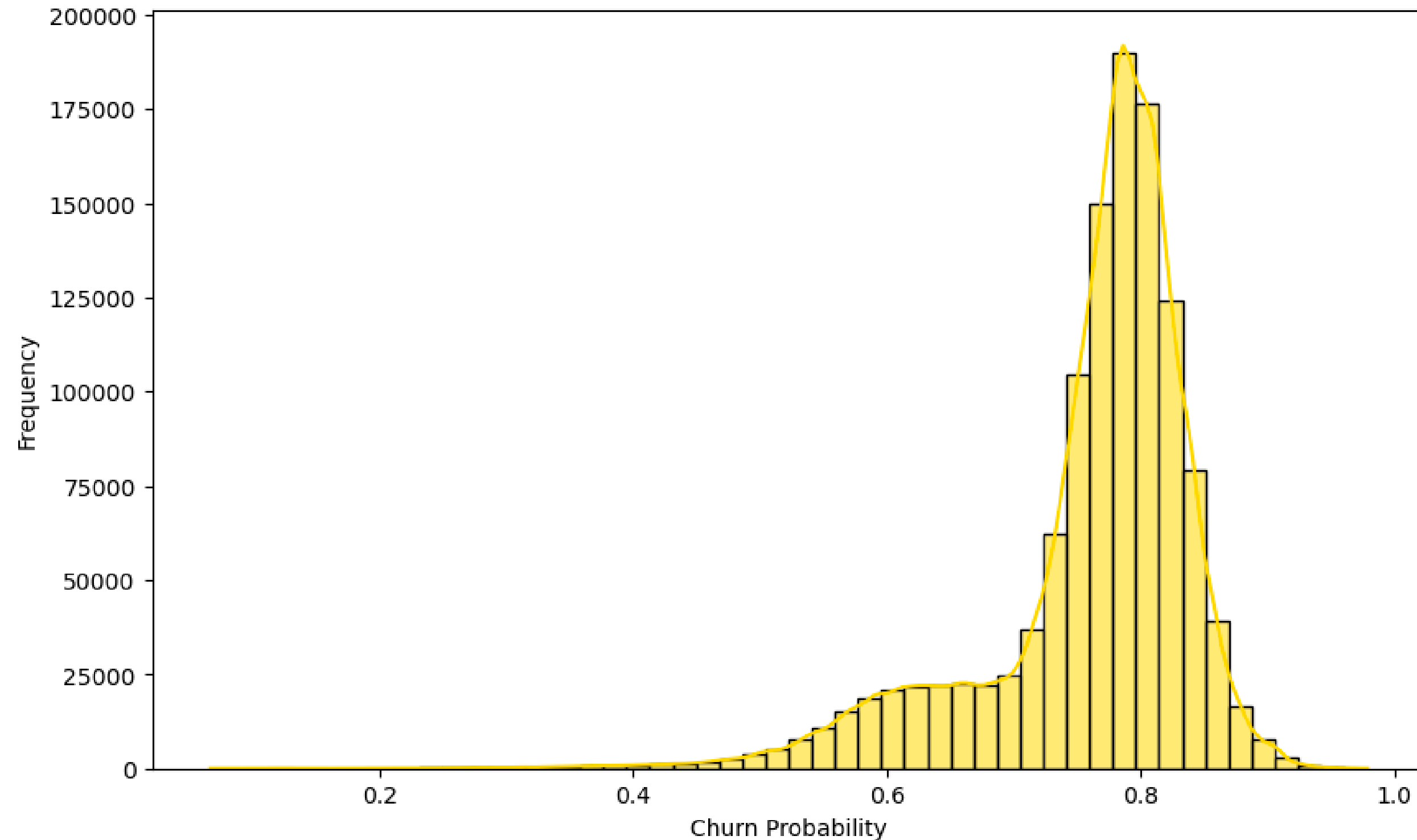


XGBoost model

Accuracy score -
76.48%



Histogram of Churn Probabilities



Distribution of Risk Segments

